DataAnalyssts

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Theme 2: Customer Retention & Telecom Analytics

Our client: JB Link,
a small Californian telecom company

Problem statement:
How might we maintain JB Link's **competitiveness in California's telecom market** so as to **reduce churn rate** due to customers switching over to competitors' offerings?

Key objectives:

- Analyse data to understand drivers of churn
- Build predictive models to identify customers at risk
- Recommend personalized actions to retain customers

Steps taken for analysis:

1. Data cleaning
2. Exploratory data analysis
3. Quantitative exploration
4. Qualitative exploration
5. Business-level analysis
6. Development of strategies

Quantitative analysis

1. **Briefly describe your data cleaning process**
   We focused on cleaning blank values by replacing them with null values in the dataset so as to not interfere with the analysis, especially the modelling. We also checked through for any outliers and found that most of them lie in the 'Populations' category which is insignificant as population spread is bound to be irregular, thus we ignored these outliers. The other categories are relatively consistent and thus we assumed no outliers.

2. **What is the median tenure of churned customers and the median tenure of non-churned customers?**
   10 years for churned customers, 38 years for non-churned customers

3. **What are the top 5 cities with the highest churn rates? What is the most common reason for churn for the 5 cities, respectively?**
   The top 5 cities by churn rate are San Diego, Los Angeles, San Francisco, San Jose, and Fallbrook. The most common reason for churn is 'Competitor made better offer', 'Competitor had better devices', 'Competitor had better devices', 'Competitor had better devices' and 'Competitor made better offer' respectively

4. **Which customer segments have the highest churn rates based on demographics, location, or service usage?**
   Women, customers aged 44, 47 and 53, unmarried customers, zip code area 92122, and the internet service type are the customer segments that had the highest churn rates

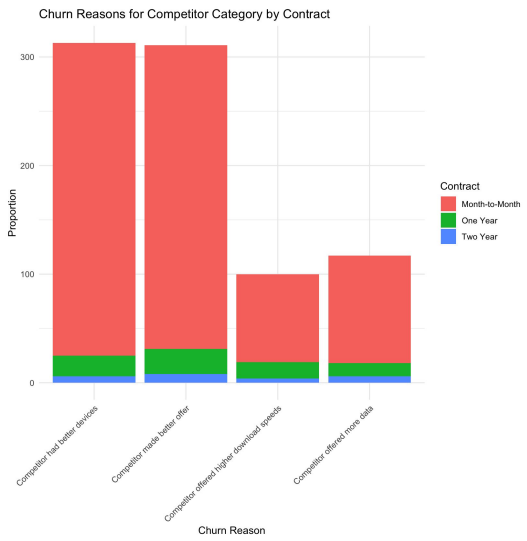
4. **What are the distributions of the payment methods adopted by customers?**
   55.5% by bank withdrawal, 39.0% by credit card, 5.5% by mailed check

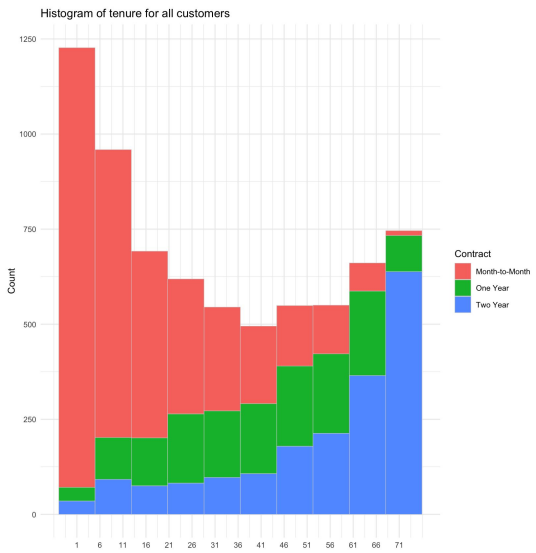
Variable	Variable type
Tenure in months	Continuous Variable
Payment method	Categorical Variable
Monthly charge	Continuous Variable
Gender	Categorical Variable
Age	Continuous Variable
Churn value	Categorical Variable
Customer satisfaction	Discrete Variable within the dataset, but can be treated as categorical or continuous data

EDA

1. **Most of their customers have month-to-month contracts and tenure of less than one month**
   Looking at the histogram of tenure for all customers, we see that a significant portion of customers have a tenure of 12 months and lower. Majority of customers also have month to month contracts (51.3%), especially concentrated at the shorter tenures (0-2 years).

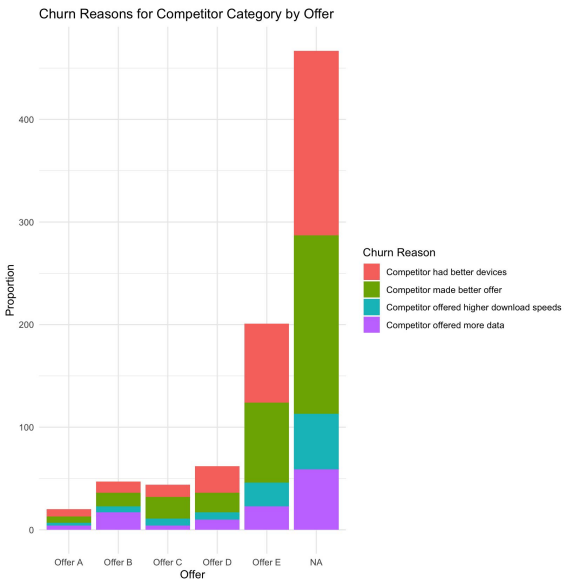
2. **Early focus on Competitors**
   When focusing on the customers that churned, We realised that it was the largest category of reasons (45%) that the customer churned. Within the category the reasons stated were that other companies had better devices (37.2%), better offers (37%), offered higher download speeds (11.9%) and offered more data (13.9).

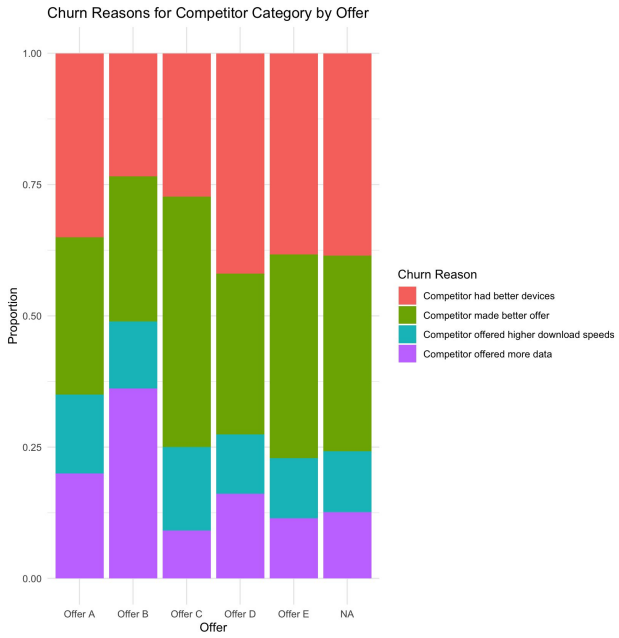




EDA

1. **Offers could be a reason to churn**
   Those without offers are the most likely to churn. The second most likely to churn is those who have offer E. When breaking down the reasons to churn for those with no offers or offer E, we find that most of the time, companies offered better devices or better offers.

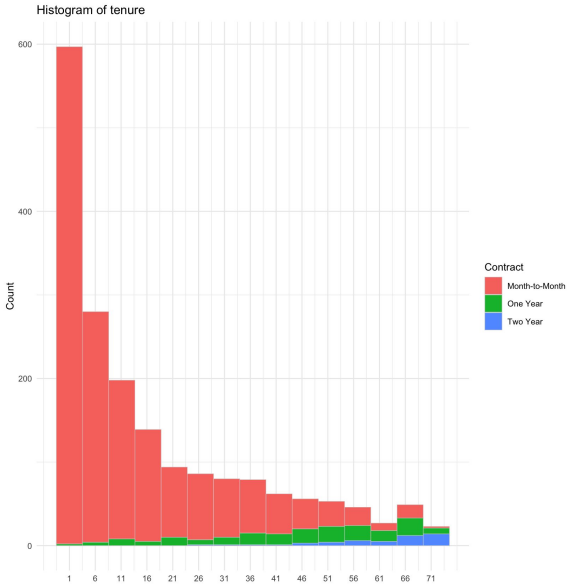


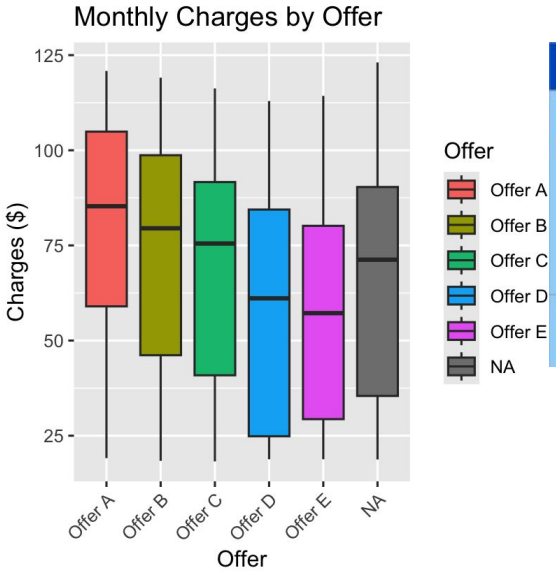


EDA

1. **Most churn happens within the first year of joining, heavily driven by flexible (month-to-month) contracts.**
   The histogram shows most customers leave within the first few months, with a high spike at very low tenure values. This early churn is almost entirely from month-to-month contracts, while one-year and two-year contracts have much steadier, lower churn rates across all tenure periods.

2. **Pricing is a factor relative to other companies' pricing plans.**
   The boxplot shows the median pricing of the different offers by JB Link, which have been compared against other telecom service providers within the region. As it can be observed, most of the offers are more pricey than the offers provided by other telecom services.





	JB Link	Spectrum	AT&T
Plan types and monthly charge	5 plans, A ($77), B ($71), C ($67), D ($58), E ($57) No offer, just add-ons ($65)	3 internet plans: Budget ($30), Family ($50), Large homes ($70) 1 unlimited data phone ($30)	1 internet plan: $60 3 mobile plans: Starter ($35), Extra ($40), Premium ($50)
Devices/uses supported	Phone, Internet	Phone, Internet, Voice	Phone, Internet

Logistic Regression Model

Features	Coefficient
Total customer service requests	+22.544
Tenure in Months	-18.483
Monthly charge	+16.207
Product/Service Issues Reported	+12.233
Number of Referrals	-9.283
Number of Dependents	-6.689
Total Extra Data Charges	+6.167

Confusion Matrix

```
        model
actual     0    1
     0  1455   97
     1   197  364
```

Technical steps:

- Dataset not too proportionate (0.730 vs 0.27) → Hence, Stratified sampling was used
- Backward elimination for significant features
- VIF → no multicollinearity
- 70-30 Train-test-split

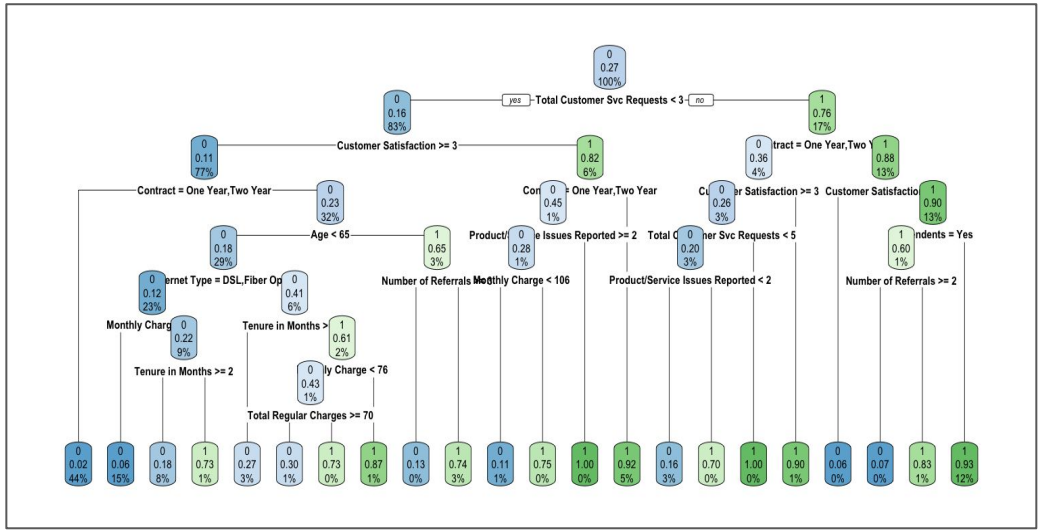
Model Results

Model Complexity	False Positive Rate	False Negative Rate	Overall Error	Accuracy
7	0.0622	0.3515	0.1391	0.8609

Business Insights from Logistic Regression Model

- High service requests, Monthly charges, Extra Data Charges, Issues Reported signaling customer dissatisfaction
  → *Facing **more problems and costs**, hence more likely to churn because **dissatisfied***
- Loyal, dependent, and referral-driven customers show stronger brand loyalty and retention
  → ***Sustained engagement and advocacy** (from referral), hence less likely to churn because of **lock in- effect and positive strong brand connection from referrals***

CART Model



METHOD

This CART model used the **Gini impurity** criterion to predict customer churn. It was trained with minsplit = 2 and cp = 0 to allow full tree growth, then **pruned using cross-validation** to avoid overfitting. Key predictive variables identified include **Total Customer Service Requests**, **Customer Satisfaction**, **Contract Type**, **Tenure in Months**, **Monthly Charges**, and **Number of Referrals**. The model outputs clear, interpretable decision rules and churn probabilities.

Confusion Matrix

	cart.predict	
Actual	0	1
0	1463	89
1	136	425

Key Business Insights

1. **<u>Service Quality is the Strongest Retention Driver</u>**
   Customers with fewer service requests (< 3) and higher satisfaction (≥ 3) show the lowest churn rates, emphasizing the importance of proactive issue resolution and responsive customer support.
2. **<u>Contract Type and Tenure Influence Loyalty</u>**
   Long-term contracts (One/Two-Year) and longer tenures (≥ 2 months) are linked to reduced churn, suggesting that promoting contract renewals and loyalty programs can strengthen retention.
3. **<u>Pricing and Engagement Impact Churn Risk</u>**
   High monthly charges and low engagement (few referrals or dependents) correlate with higher churn. Optimizing pricing tiers and incentivizing referrals can improve customer stickiness.

Model Comparison

Logistic Regression

Key Predictors: Total Service Requests (+), Monthly Charges (+), Product/Service Issues Reported (+), Tenure in Months (–), Number of Referrals (–)

Performance:

- Accuracy: **86.1%**
- False Negative Rate: **35.1%**
- False Positive Rate: **6.22%**

CART

Key Predictors: Total Service Requests, Customer Satisfaction, Contract Type, Monthly Charges, Tenure in Months, Number of Referrals

Performance:

- Accuracy: **89.4%**
- False Negative Rate: **24.2%**
- False Positive Rate: **5.73%**

Logistic Regression provides **quantifiable impact** for strategic prioritization.
CART provides **intuitive decision rules** for operational implementation and targeting.
Together, they offer **complementary insights**. The statistical rigor of LR and the interpretability of CART enhance decision confidence.

Key business findings

1. Competitor Influence & Pricing Pressure

- Most customers cite competitors as their reason for leaving, especially due to cheaper plans and better devices.
- JB Link offers a wider variety of plans than competitors (AT&T, Spectrum), but their prices remain higher than the lowest competitor options.
- Insight: Pricing and contract models make switching easy, particularly for short-term/month-to-month customers.

2. Contract Type & Tenure Drive Loyalty

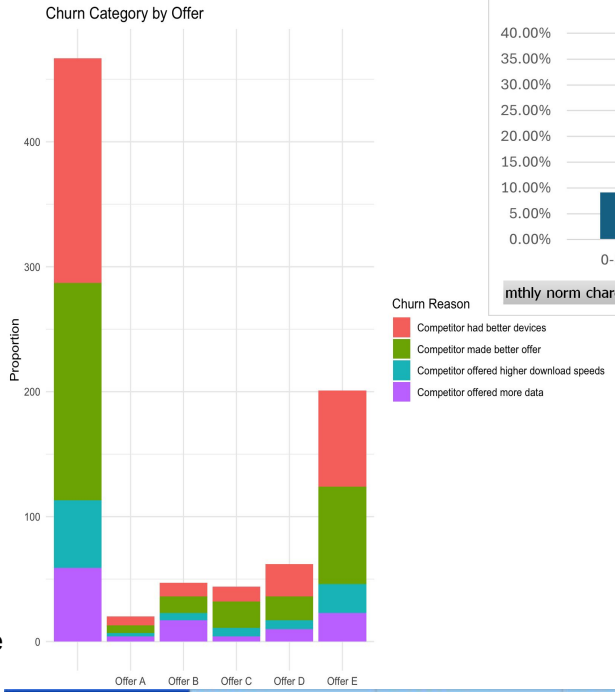
- Long-term contracts (1–2 years) and longer tenure reduce churn.
- Insight: Encouraging renewals and offering loyalty incentives strengthens retention.

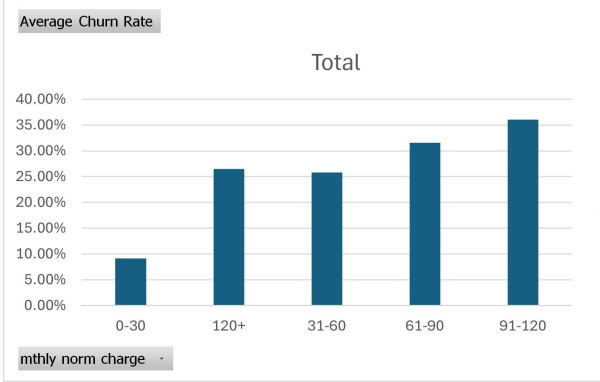
3. Service Quality is the Strongest Retention Driver

- Fewer service requests (<3) and higher satisfaction (≥3) significantly reduce churn.
- Low engagement (few referrals/dependents) and high service requests indicate customer dissatisfaction.
- Insight: Proactive issue resolution and responsive customer support are critical to retaining customers.

4. Retention Campaign Opportunities

- High-risk segments: short-tenure, high monthly charges, low engagement.
- Studies show that anti-churn campaigns (e.g., targeted promotions, personalized support) have proven positive effects on retention.





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Recommended strategies

1. Competitive & Flexible Pricing Model

- Why: Customers are leaving for cheaper competitor plans and devices.
- Introduce **tiered pricing and affordable entry-level plans** to match competitor offers.
- Offer discounted plans that are automatically **triggered when a customer's churn** likelihood **exceeds a certain threshold** (via a predictive model)
- **Implement dynamic pricing or "price match"** options for high-risk, short-term users.

2. Strengthen Loyalty Through Contracts & Rewards

- Why: Longer contracts and tenure significantly reduce churn.
- Promote **1–2 year contract renewals** and **conversion of short-term contracts to 1-2 year contracts** with l**oyalty rewards** (e.g., bill credits, extra data, or premium features).
- Launch a **tiered loyalty program** (bronze, silver, gold) that offers increasing benefits with tenure (e.g., VIP support or upgrade priority).
- Encourage **contract upgrades** before expiry through **personalized renewal incentives**.

3. Service Quality Enhancement & Proactive Support

- Why: Service quality and satisfaction are the strongest retention drivers.
- Establish **proactive support systems** that identify and resolve service issues early.
- Track **service request frequency** and **flag dissatisfied customers for priority handling**.
- Invest in **training and customer experience analytics** to improve responsiveness and problem resolution speed.

4. Targeted Retention & Engagement Campaigns

- Why: Short-tenure, high-charge, and low-engagement customers are most at risk.
- Use **predictive churn models** to **target high-risk segments** with personalized offers.
- Launch **anti-churn campaigns** such as discounts, exclusive deals, or check-ins for disengaged users.
- **Incentivize engagement via referral programs** and dependent-line bundles to strengthen customer ties.

Thank you !

Appendix

https://www.indeed.com/companies/best-Telecommunications-companies-in-California

https://www.att.com/buy/internet/plans

https://www.att.com/plans/wireless/

https://www.highspeedinternet.com/resources/spectrum-mobile-deals

https://www.highspeedinternet.com/resources/spectrum-real-costs

https://www.sciencedirect.com/science/article/abs/pii/S0308596120301609